



Flow dynamics through cellular material based on a structure with triply periodic minimal surface

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ABSTRACT

Triply Periodic Minimal Surface (TPMS)-based cellular structures are widely adopted due to their high surface-to-volume area, superior mechanical characteristics, and adaptive properties. This paper is dedicated to deriving a new semi-empirical equation for pressure drop in TPMS-based structures (specifically Schwarz-P, Schoen I-WP, Neovius). Both numerical and experimental tests were conducted to analyze the impact of TPMS parameters on pressure drop and velocity distribution. It was determined that flow through the Schwarz-P structure exhibits a tubular flow pattern, flow through the Neovius structure demonstrates active flow mixing, and flow through the Schoen I-WP structure does not have stagnant (dead) zones. A novel semi-empirical equation was formulated to calculate pressure drop as a function of flow parameters and TPMS-based structure parameters. This equation incorporates both viscous resistance ($\Delta p = f(v)$) and kinetic energy loss ($\Delta p = f(v^2)$), enabling its application for predicting pressure drop across a wide range of practically interesting operational parameters.

1. Introduction

Porous materials are widely used due to their numerous advantages (Su et al., 2023; Igwe et al., 2023; Dzanic et al., 2023). Porous materials are used in filter manufacturing as they have high surface area and variable permeability (Chen et al., 2023). Bone implants are created from porous material and are very compatible with the human body (Alvarado-Hernández et al., 2023). Light and strong mechanical components used in the aerospace industry are also manufactured from porous materials (Pramanik and Manna, 2022). Their use is also widespread in battery production (Siddique et al., 2022), agricultural (Ma et al., 2022) and chemistry industry (Raghunath et al., 2022), energy sector (Sheikholeslami, 2022) and so on (Qin et al., 2022; Habibishandiz and Saghir, 2022; Zeng et al., 2023; Zhan et al., 2023).

Despite the everyday use of porous materials and their multiple advantages, they also have disadvantages: pores are easily clogged, high pressure drop, poor mechanical properties, heterogeneous pore distribution, etc. Channel clogging causes lesser permeability, leading to decreased volumetric fluid flow. Higher pressure drop leads to increased strain on the pump and higher power consumption. Poor mechanical properties reduce structure service time and limit its use. The porous materials with ordered structures are capable of solving these issues (Baobaid et al., 2022; Lee et al., 2022). A lot of research in the last decade has been dedicated to their study.

Triply periodic minimal surfaces (TPMS) are surface-type cellular structures that lower the strain rate, reduce the structure's weight and control heat transfer characteristics (Eremin et al., 2024). The main advantage is the flexibility of their use due to adjusting their main design parameters: porosity and cell size. High porosity increases the structure's surface area, and low porosity is used to improve mechanical properties. TPMS cell size is varied to control the packed bed permeability or to adjust structure thermal conductivity in insulation applications. Also, a wide choice of various TPMS structures allows for selecting effective cell types for each problem (Li et al., 2022; Hesselmann et al., 2022; Ryan, 2016; Sreedhar et al., 2018).

TPMS packed beds have good heat transfer characteristics and high specific surface area. Also, the flow inside these packed beds is divided into two non-intersecting zones: internal and external (Benjamin et al., 2023). Thus, TPMS packed beds could be effectively used in heat exchangers (Alteneiji et al., 2022). This type of heat exchanger is not yet used in industry since the flow and pressure drop characteristics haven't been thoroughly studied.

The latest research suggests that using TPMS in bone tissue engineering provides numerous advantages (Viet et al., 2023). Inside the bone tissue occurs a constant flow of blood and marrow. To efficiently use TPMS as bone implants, the permeability and wall shear stress influence on the flow require further study.

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The high specific surface area of TPMS is well-suited for use as the coated catalyst surface. The coated catalyst surface is used in fixed bed catalytic reactors (Pashchenko, 2023). Among the key parameters of such a packed bed are pressure drop and contact area (Karpilov and Pashchenko, 2023). It is crucial to study pressure drop and flow dynamics to use the TPMS-based packed bed geometry in chemical reactions.

The hydrodynamic properties of the flow inside TPMS packed beds are still being studied extensively due to the reasons stated above. Pires et al. numerically and experimentally investigated various TPMS geometries for bone tissue engineering (Pires et al., 2021). Three TPMS geometries were considered: Schwarz D, Gyroid and Schwarz P. Each geometry was used twice to create two structures with porosity of 60% and 70%. The authors obtained power dependence of pressure drop on volumetric flow rate and determined that higher porosity does not always lead to higher permeability.

Singh et al. studied flow dynamics in a cylindrical column filled with TPMS-based packed bed (Singh et al., 2022). The column is designed for carbon capture on a wet surface. To create a packed bed, the Schwarz-D structure is used. A numerical and experimental study of pressure drop, flow regime in a column under various gas and fluid flow rates, contact angle, solvent properties and interfacial area was carried out. The authors obtained a semi-empirical dependence of pressure drop on column height and gas load. Also, under high velocities with a high gas flow rate, an unstable wet flow was observed and its pressure drop was measured.

Hydraulic and heat transfer properties of heat exchangers with TPMS Schoen-G and printed circuit heat exchanger (PCHE) structures were compared by Li et al. (2022). The streamlines in the TPMS Schoen-G heat exchanger are considerably smoother downstream, even though a higher velocity appears in the areas with greater curvature. The streamlines near the wall are trapped in each corner in PCHE, and spiral flow patterns are observed, which led to a significant hydraulic loss but heat transfer enhancement. The novel TPMS Schoen-G heat exchanger showed higher heat transfer properties than PCHE for a given inlet temperature and pressure boundary conditions.

Khalil et al. numerically examined the thermal and hydraulic properties of the heat sink with TPMS Diamond-D and Gyroid-G structures (Khalil et al., 2022). Heat exchangers had porous elements with a constant cell size of 10 mm and porosity of 80%. Authors reported that in both geometries, the dependence of friction factor on Reynolds number was linear and Diamond-D friction coefficient was four times lower than for Gyroid-G. Gyroid-G geometry was more efficient as pressure drop per unit length was 20% lower than for Diamond-D. However, thermal performance was better for Diamond-D geometry as the heat transfer coefficient was 26% higher than for the Gyroid-G structure.

Tisha et al. created a liquid-to-liquid compact heat exchanger with a gyroid lattice by additive manufacturing. The heat exchanger was light and compact with 80% porosity, a cell wall thickness of 0.3 mm and a surface-to-volume ratio of $670 \text{ m}^2/\text{m}^3$. The authors experimentally determined the heat transfer coefficient and Reynolds number and numerically calculated pressure distribution. The proposed heat exchanger geometry showed an efficiency increase of 55% compared to the thermodynamically equivalent countercurrent heat exchanger. Moreover, the proposed TPMS-based heat exchanger was ten times smaller than the countercurrent one. The total pressure drop was negligible for one structure cell size of $4.6 \times 4.6 \times 4.6 \text{ mm}$ but was not investigated for the whole structure.

Periodic nodal surfaces (PNS) have similar structures to TPMS, and their characteristics are deemed very promising (Iyer et al., 2022). Iyer et al. numerically explored thermal and flow characteristics of 7 TPMS and PNS heat exchanger geometries (Iyer et al., 2022). The calculations were carried out for a structure containing four consequent single-unit cells and a laminar flow regime. The authors obtained friction factor correlations for the flow in each geometry for the Reynolds number range of 10–300.

The connection between hydraulic parameters, heat transfer coefficient and structure strength was explored by Cheng et al. (2021a). The authors considered several topologies: W type, P type, D type and G type. The dependence of pressure drop per unit length on low Reynolds number (10–150) was calculated for one row with eight porous cells. The velocity streamlines inside porous media are also presented in the article. Streamlines in P-type geometry show zones with very slow velocity perpendicular to the main flow. In their subsequent research (Cheng et al., 2021b), the authors determined that TPMS structure was better than simple cubic in terms of pressure drop to heat transfer coefficient ratio and structural strength.

Cell material thickness also has a significant influence on thermal and hydraulic characteristics. Gas flow inside Schwarz D structure with varying cell thickness was investigated by Attarzadeh et al. (2021). The cell dimensions were $10 \times 10 \times 10 \text{ mm}$ and were placed in a tube as a $4 \times 4 \times 4$ cubic structure. The numerical analysis showed that pressure drop has a quadratic dependence on the cell thickness, especially for a lesser porosity. Change in porosity from 59% to 88% led to increased pressure drop by more than five times. Also, a non-linear permeability reduction was observed with the linear TPMS porosity decrement.

The uncertainty of hydraulic characteristics of TPMS-based structures limits their practical application. This paper aims to gain an understanding of the pressure drop dependence on the design parameters of TPMS-based structures. Both numerical and experimental tests were performed under various inlet velocities, structure porosities and cell geometric characteristics. The flow was analyzed through widely used TPMS-based structures such as Schwarz-P, Schoen I-WP and Neovius. The novel contribution of this work is to develop a single semi-empirical equation for estimating the pressure drop over various TPMS-based structures. This was achieved firstly by comparing the experimental and numerical results with classical equations and secondly by deriving a new semi-empirical equation. Also, the numerical models were used to determine the flow structure, which allowed for identifying flow patterns, stagnant regions, and areas of mixing between different streams.

2. Methodology

2.1. 3D printing

The following TPMS geometries were chosen for the research: Schwarz P, Neovius, and Schoen's I-WP. All three surfaces exhibit reflectional symmetry about the three mid-planes and orthogonal rotational symmetry about the three axes. Therefore, the considered cells possess cubic symmetry.

In this study, Surface Evolver was chosen for TPMS surface generation due to its extensive range of model processing tools (Poureslami et al., 2021; Ma et al., 2021; Kolken et al., 2022). A wide range of TPMS equations derived from Surface Evolver and various available TPMS makes Surface Evolver a universal tool for creating topological structures (Cheng et al., 2021b; Iyer et al., 2022; Wohlgemuth et al., 2001).

The created TPMS surface was then processed in Solidworks, after that the whole structure with multiple cells was generated. A solid layer was formed by generating two equidistant surfaces and subsequently filling the gap between them with solid material. The whole workflow of creation the TPMS structure for 3D printing is shown in Fig. 1.

The flow was investigated only inside the structure. Therefore, the flow channels on the external side were sealed. In the Fig. 2, the sealed exterior outlets are denoted as A and B. On the other two sides, the external outlets are also closed. Gas flow occurs only through the internal space of the TPMS cells as the cell's outer channel is sealed.

The sample CAD geometry is imported into the "Chitubox Basic" software. Settings were kept at the default values according to the manual provided by Phrozen (2023a). The technology chosen for sample printing was LCD (Liquid Crystal Display), which allows for high-precision fabrication. The Phrozen Sonic Mighty 4K printer was used for the 3D printing. Phrozen Aqua Gray 4K and Phrozen Aqua Blue were

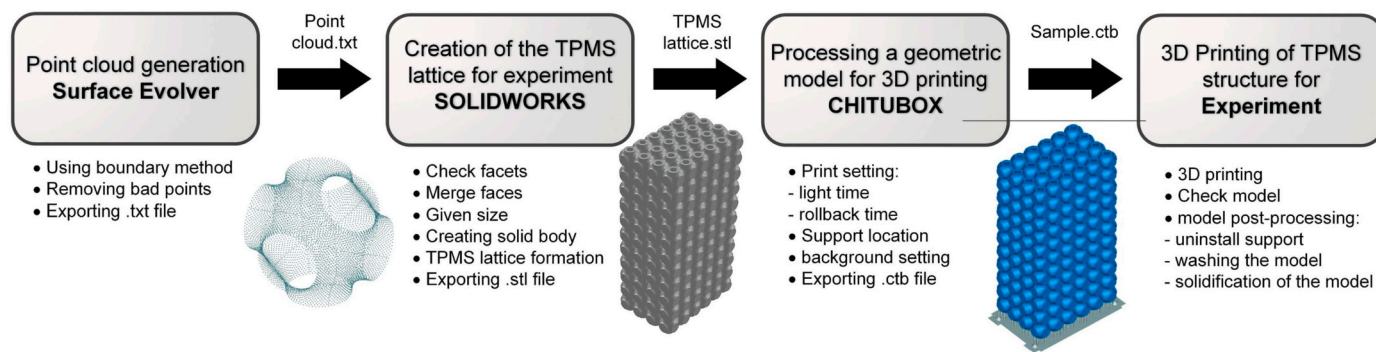
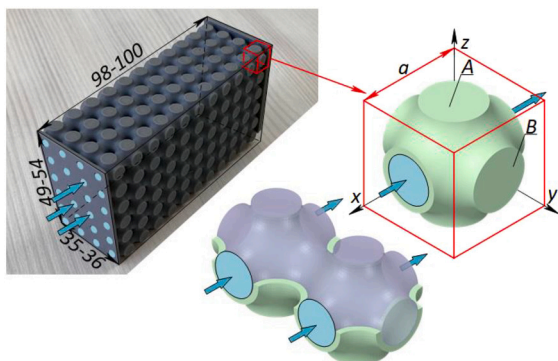


Fig. 1. Step by step TPMS generation.

Fig. 2. Printed Schwarz-P TPMS packed bed with cell size $a = 7$ mm. A and B are sealed external outlets.

selected as polymer materials. The material's properties were provided by the company and are available on the official web page (Phrozen, 2023b).

After printing, the model was thoroughly washed in a special liquid to remove any residual photopolymer resin. The rinsed sample was then exposed to ultraviolet light for the final curing of the model.

2.2. Experimental setup

The experimental setup for pressure drop investigation was used to validate the findings. The experimental setup is shown in Fig. 3. It includes an air pre-treatment section, a packed bed section and measuring equipment. The air pre-treatment section consists of a fan for the flow motion and an air heater that heats air to the defined temperature. The packing bed section includes a tube with an experimental sample (TPMS structure) inside. Measuring equipment controls the flow characteristics and ensures similar operating conditions.

The experimental setup consists of the following equipment: air intake, axial fan (Ostberg CK 200 A), honeycomb flow straightener, air heater (PBEC 200/2.2), air duct, square-to-round duct adaptor, packed bed, differential pressure manometer (Regin MV600), a pitot tube and a control panel.

An axial fan creates air inflow from the laboratory room to the duct. Then, a honeycomb flow straightener is used to suppress the axial flow component. Thus after the straightener, the flow is fully developed with no additional axial velocity. To ensure similar air viscosity and density, the flow is heated in the air heater to 28°C . After the heater air moves through a cylindrical duct up to the square-to-round duct. In the square duct the TPMS-based structure is placed. The distance between the TPMS sample and the square-to-round duct is 400 mm to exclude the flow transition influence. When the airflow passes through a packed bed, it loses pressure due to the high local resistance of the TPMS sample. A differential pressure manometer measures the pressure drop. The flow velocity is measured by a pitot tube near the duct

outlet. The differential manometer and pitot tube are connected to a control panel that transforms data into a digital format and shows it on the display.

A set of experiments was conducted to collect data for various boundary conditions. The fan speed varied between 960 - 2820 rpm, changing the airflow velocity from 1 to 2.5 m/s. The TPMS structures were TPMS-P geometries with different cell sizes: 5 mm, 7 mm and 9 mm, which are presented in Fig. 2. To sustain equal packed bed length, the number of cells was varied from the cell size as shown in Table 1. Each experiment was repeated five times under the same operating and boundary conditions to determine the uncertainty range.

2.3. Numerical modeling

2.3.1. Geometry

The numerical simulation of fluid flow in the TPMS structure was performed using the Ansys Fluent 19R2 software. The geometry creation procedure for CFD modeling was similar to one used for obtaining the experimental samples.

The whole process of geometry creation for CFD modeling is shown in Fig. 4. The initial surface S1 (Fig. 4a) is an imported TPMS surface. This surface of the TPMS cell is bound by a cube with the edge size of a mm (Table 1), as shown in Fig. 4a.

Then the two equidistant surfaces S2 and S3 were built on a distance equal to $\delta/2$ from the initial surface S1. A space is created between them that is filled with material, thus creating a solid wall. After that the cube with the edge size of a mm was created and a total of three bodies are present (Fig. 4b): Cube - volume of the external part of the cube, by which the unit cell is bounded; Fluid - inner region of the cell, where the fluid flow occurs; Lattice - volume body of the cell wall. The inner region of the cell (Fluid) was subtracted from the combined volume of the Cube and Lattice. Thus, the cell with only fluid volume was generated as shown in Fig. 4(c).

The whole TPMS structure for Schwarz-P unit cell of 7 mm (Fig. 4 c) has the following dimensions in X, Y and Z directions: 98 mm, 49 mm and 35 mm as shown in Fig. 4(d). Additional zones were added before and after the TPMS structures to form a developed flow and dissipate the eddies before the fluid outlet.

According to the algorithm described above geometries of unit cells of the Schwarz P, Schoen's I-WP and Neovius TPMS were obtained.

2.3.2. Mesh generation procedure

The mesh for the Schwarz-P geometry is generated in the Ansys Fluent meshing module and is presented in Fig. 5. The mesh is composed of polyhedral elements with a boundary layer near the cell wall. The boundary layer is formed by the layered prismatic elements. The total number of layers was 9 with a first layer thickness of $1.44e^{-5}$ m and layer stretching of 1.2. The mesh quality was checked using several parameters: aspect ratio, skewness and orthogonal quality. Maximum and average values for aspect ratio are 28.76 and 3.64; for skewness 0.899 and 0.197. Minimum orthogonal quality was 0.13 and 0.887 on

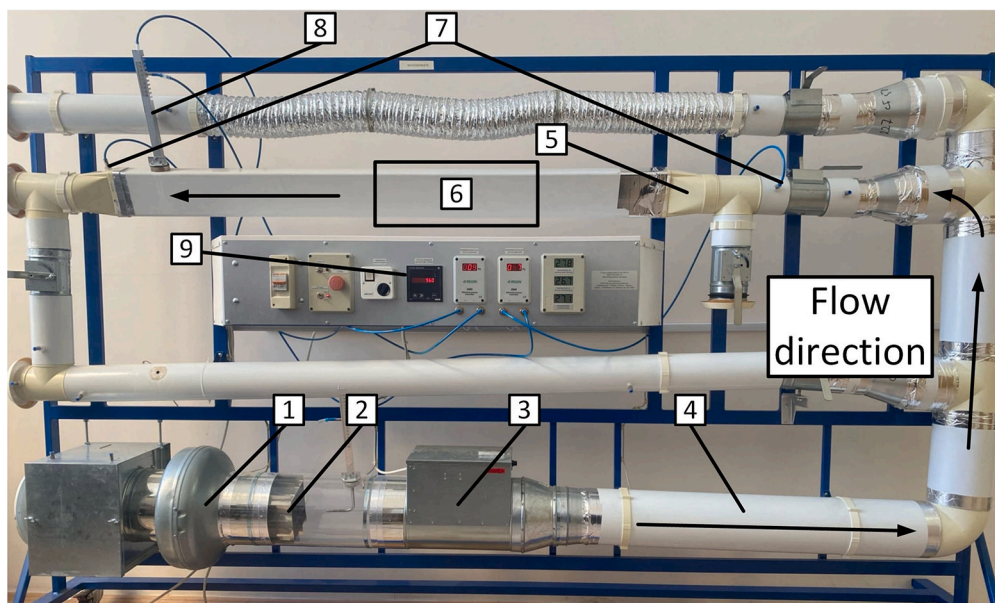


Fig. 3. Experimental setup for pressure drop measurement. The following equipment is shown: 1) axial fan Ostberg CK 200 A; 2) honeycomb flow straightener; 3) air heater PBEC 200/2.2; 4) air duct; 5) square to round duct adaptor; 6) packed bed; 7) differential pressure manometer Regin MV600; 8) pitot tube; 9) control panel.

Table 1
3D printing settings Schwarz P.

Size a (Fig. 2), mm	Thickness δ , mm	Number of cells in X, Y and Z direction (Fig. 2)	Structure dimensions in X, Y and Z direction (Fig. 2), mm	Porosity ϵ
5	0.6	20×7×10	100×35×50	0.35
7	0.85	14×5×7	98×35×49	0.35
9	1.09	11×4×6	99×36×54	0.35

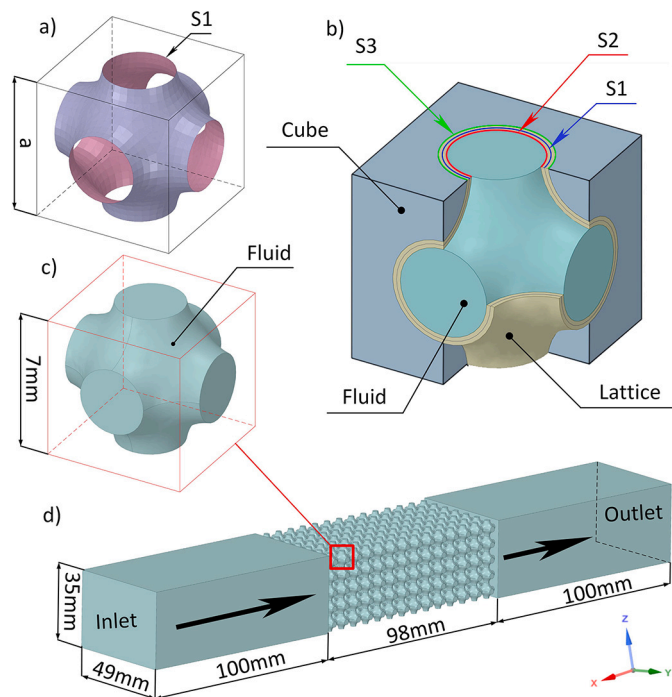


Fig. 4. Computational geometry: a) imported surface; b) initial cell; c) cell geometry used for CFD; d) whole TPMS structure.

average. All the parameters are in the acceptable range, according to Park et al. (2022). These settings were chosen according to the con-

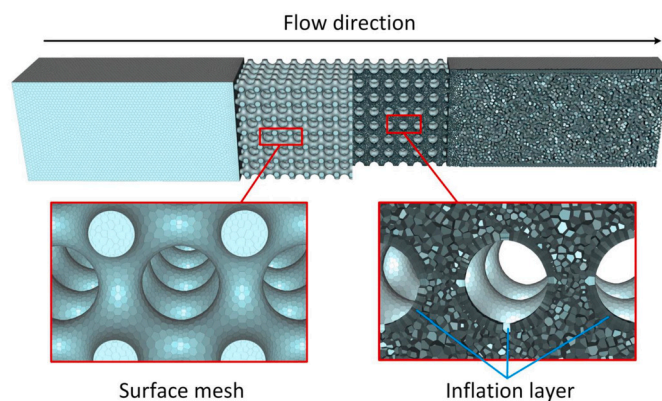


Fig. 5. Mesh of the Schwarz-P model.

dition: $y^+ < 1$. Total number of elements was 2906576 for the fluid region.

The mesh sensitivity analysis was performed to measure the accuracy of the simulation with respect to the number of mesh cells. Five mesh sizes were adopted and compared for each TPMS geometry. The TPMS-P mesh consisted of the following elements: 1488166, 1860208, 2325260, 2906576 and 4208969. The pressure drop acquired on the finest mesh was compared to the other meshes and the percentage difference was evaluated. The mesh independence study results are presented in Fig. 6. The pressure drop deviation from the finest mesh of 4208969 elements is 10.66%, 6.2%, 2.44% and 0.49% for 1488166, 1860208, 2325260 and 2906576 elements, respectively. The mesh with 2906576 elements is accurate enough as the deviation from the finest mesh is less than 1%. Therefore, the mesh with 2906576 elements can

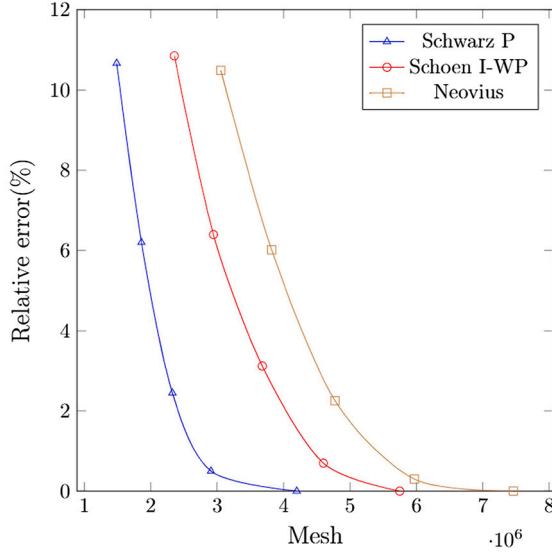


Fig. 6. Average pressure drop for different TPMS structures on various mesh.

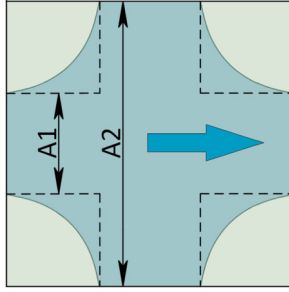


Fig. 7. Schematic model of a fluid flow inside porous media.

save computational time and is adopted for the present investigation. For Neovius and Schoen's I-WP TPMS packed bed geometries, the mesh sensitivity analysis was carried out similarly and its results are presented in Fig. 6.

2.3.3. Governing equations

In the present article flow properties of TPMS packed bed are investigated via numerical simulation in Ansys Fluent 2019 R3 software. The airflow inside the TPMS structure is turbulent due to the high Reynolds number and constant change of the flow channel cross-section, which causes the air velocity to increase significantly. Turbulent flow inside various TPMS structures may be simulated using several turbulence models. One commonly used turbulence model is $k - \omega$ SST due to its accurate pressure drop prediction (Alatyar et al., 2023; Khalil et al., 2022; Subasi et al., 2017; Papkov et al., 2023, 2024).

To sustain equal mass flow on the inlet and outlet of the model, the continuity equation is solved for a steady state flow:

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0 \quad (1)$$

The momentum conservation equation in Cartesian tensor form is written as:

$$\frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \mu \nabla^2 u_i + \frac{\partial}{\partial x_j}(-\rho \overline{u'_i u'_j}) \quad (2)$$

In this model, the heat transfer properties of TPMS structures were not investigated. However, the energy equation was used to implement air properties (viscosity and density) with values corresponding to the operating temperature of 301 K. The energy equation for a turbulent flow has the following form:

$$\frac{\partial}{\partial x_i}[u_i(\rho E + p)] = \frac{\partial}{\partial x_j} \left(k_{\text{eff}} \frac{\partial T}{\partial x_j} + u_i(\tau_{ij})_{\text{eff}} \right) \quad (3)$$

2.3.4. Boundary conditions

Operating and boundary conditions are listed in detail in Table 2. The calculations for each geometry were performed for uniform inlet velocities from 1 to 2.5 m/s and the pressure outlet was specified on the geometry outlet. The air temperature was set as 28 °C and kept constant as no heat exchange was present. Air density changed from pressure according to the ideal gas law. The operating pressure was set according to the atmospheric pressure. The coupled algorithm was implemented for pressure-velocity coupling and a second discretization order was set for all derivatives. The calculation was stopped after the convergence criteria of 10^{-6} was met or the pressure outlet value remained constant for an extended period of time.

2.4. Analytical approach

The pressure drop in porous media is caused by viscous and kinetic energy loss. The most widespread model for such a flow is the Ergun equation. The Ergun equation is written in the following form:

$$\frac{|\Delta P|}{L} = \frac{150\mu}{D_p^2 \phi^2} \frac{(1-\epsilon)^2}{\epsilon^3} v_s + \frac{1.75\rho}{D_p \phi} \frac{(1-\epsilon)}{\epsilon^3} v_s^2 \quad (4)$$

The superficial velocity is defined by:

$$v_s = \bar{v}\epsilon \quad (5)$$

The first term describes viscous resistance mostly encountered in laminar flow. This term is called the Blake-Kozeny equation and is written below:

$$\frac{|\Delta P|}{L} = \frac{150\mu}{D_p^2 \phi^2} \frac{(1-\epsilon)^2}{\epsilon^3} v_s \quad (6)$$

The second term of the Ergun equation calculates kinetic energy loss normally present in turbulent flow. This term is known as the Burke-Plummer equation and has the following form:

$$\frac{|\Delta P|}{L} = \frac{1.75\rho}{D_p \phi} \frac{(1-\epsilon)}{\epsilon^3} v_s^2 \quad (7)$$

Another method for pressure drop calculation in porous media is the Hagen-Poiseuille equation, which is written below:

$$\Delta P = \frac{8\mu L \bar{v}}{r^2} \quad (8)$$

Hagen-Poiseuille equation describes the pressure drop for fluid flow in a tube. Therefore, different variations of this equation are proposed to make it applicable for pressure drop calculations in a porous media (Lan et al., 2022; Lanetc et al., 2022; Sankar et al., 2022; Wu et al., 2008).

For the present investigation, a method described by Wu et al. (2008) was implemented to modify the Hagen-Poiseuille equation for the pressure drop calculation on Schwarz-P TPMS geometry. This method could be used if several assumptions are made: solid phase in porous structure is represented by spheres, and the flow shape could be represented as capillary. Therefore, the method only applies to the pressure drop calculation of a packed bed with solid particles resembling a sphere. With the given assumptions, Schwarz-P geometry is deemed the most suitable for analytical investigations.

Therefore, the simplest Schwarz-P geometry and similar but more complex - Schoen's I-WP and Neovius TPMS were represented as a sphere with equivalent volume. The particle diameter of this sphere is calculated as:

$$D_p = \sqrt[3]{\frac{6V_{\text{solid}}}{\pi}} \quad (9)$$

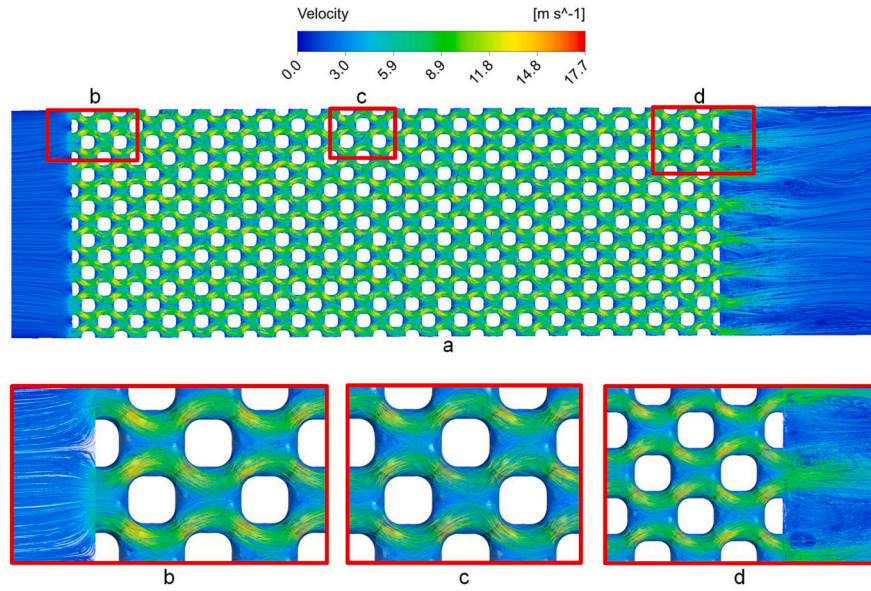


Fig. 8. The side view of flow distribution inside Schoen's I-WP TPMS structure for inlet velocity 2 m/s, porosity 0,35 and cell dimension 5 mm: a) at the flow entry inside TPMS; b) at the central part; c) at the outlet.

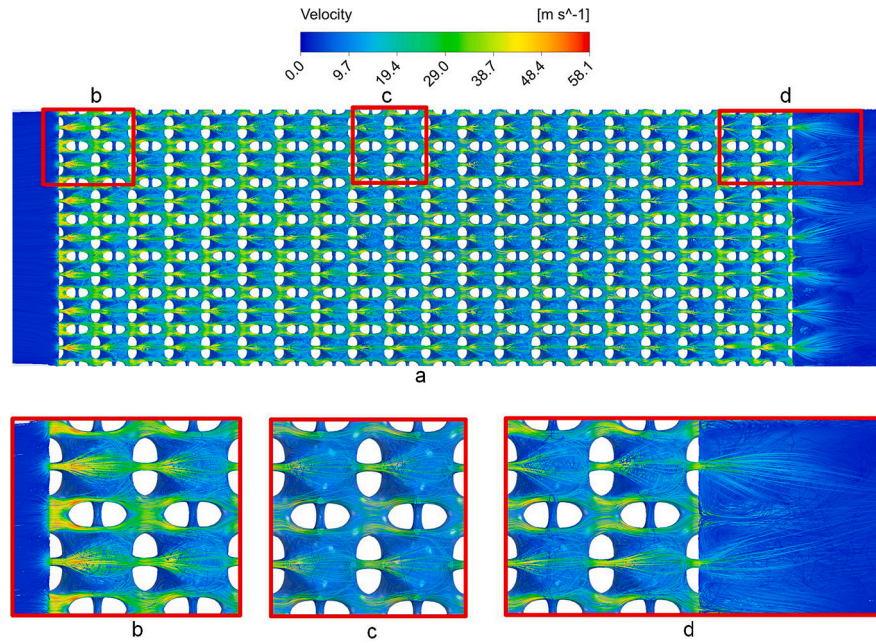


Fig. 9. The side view of flow distribution inside Neovius TPMS structure for inlet velocity 2 m/s, porosity 0,35 and cell dimension 5 mm: a) at the flow entry inside TPMS; b) at the central part; c) at the outlet.

Table 2

Boundary and operating conditions.

	Inlet velocity	Outlet	Air temperature	Air density	Air kinematic viscosity	Pressure
Value	1... 2.5	Pressure outlet	28	1.145	$13.95e^{-6}$	100247
Units	m/s	(-)	°C	kg/m ³	m ² /s	Pa

The average flow velocity was calculated using the following expression:

$$\bar{v} = \frac{q}{A} = \frac{\Delta P_1 d^2}{32\mu L \tau} \quad (10)$$

where q is the volumetric flow rate, m³/s; A is the average cross-sectional area, m²; d is the mean flow diameter, m; τ is tortuosity.

The pressure drop caused by sudden expansion and contraction of the flow in porous media is generally written in from (Wu et al., 2008; Zhang et al., 2022):

$$h_{f2} = (\xi_e + \xi_c) \frac{\bar{v}^2}{2g} = n \cdot \left(\frac{3}{2} + \frac{1}{\beta^4} - \frac{5}{2\beta^2} \right) \frac{\bar{\rho} \bar{v}^2}{2} \quad (11)$$

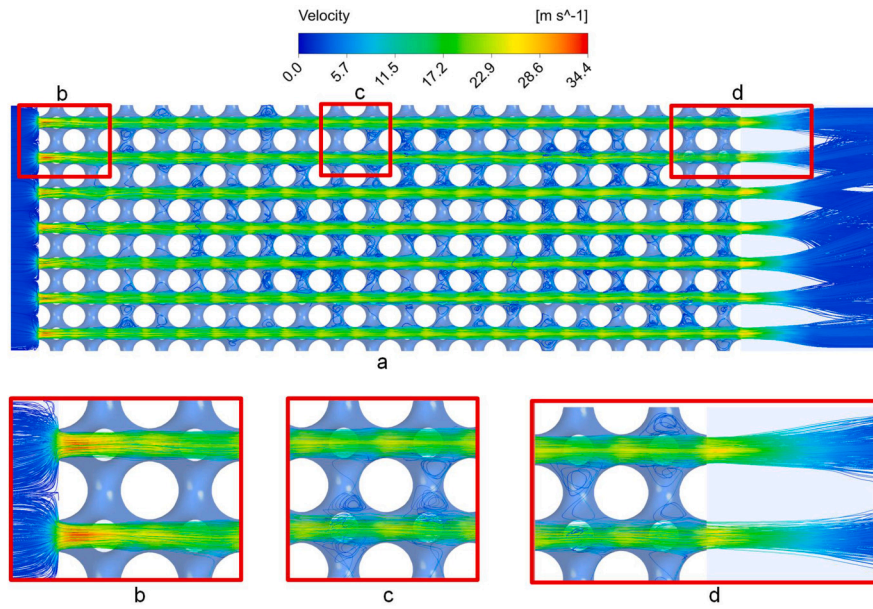


Fig. 10. The side view of flow distribution inside Schwarz-P TPMS structure for inlet velocity 2 m/s, porosity 0,35 and cell dimension 5 mm: a) at the flow entry inside TPMS; b) at the central part; c) at the outlet.

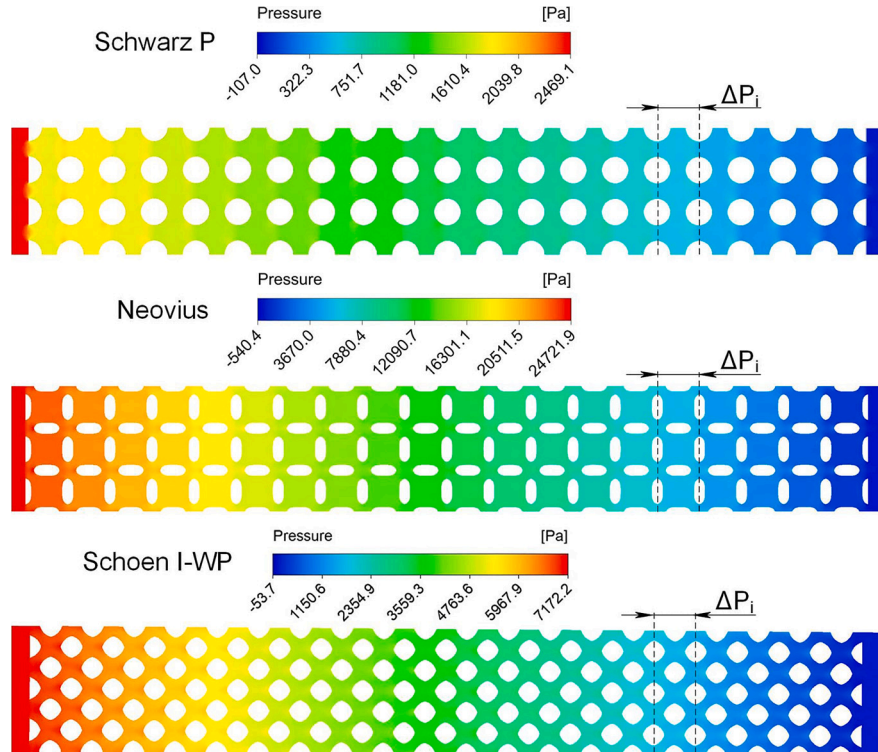


Fig. 11. Pressure drop along the packed bed length for inlet velocity 2 m/s, porosity 0,35 and cell dimension 5 mm: a) Schwarz-P, b) Neovius and c) Schoen's I-WP TPMS geometry.

where n is a number of channels and β is a geometric characteristic defined as:

$$\beta = \sqrt{\frac{A_2}{A_1}} = \frac{A_1 + D_p}{A_1} \quad (12)$$

A_1 and A_2 are the diameter of the narrow and wide channels as shown in Fig. 7.

The sphericity coefficient ϕ of the solid particle was introduced to compensate for the effect of non-spherical particles on the flow.

The Hagen–Poiseuille equation (8) could be rewritten with equations (9)–(12) as:

$$\frac{\Delta P}{L} = \frac{72\mu\tau(1-\varepsilon)^2v_s}{D_p^2\phi^2\varepsilon^3} + \frac{3\tau(1-\varepsilon)\rho v_s^2 n(\frac{3}{2} + \frac{1}{\beta^4} - \frac{5}{2\beta^2})}{4\varepsilon^3 D_p \phi} \quad (13)$$

3. Results and discussion

This section presents the fluid flow dynamics inside porous TPMS structures based on Schwarz-P, Neovius and Schoen's I-WP topology.

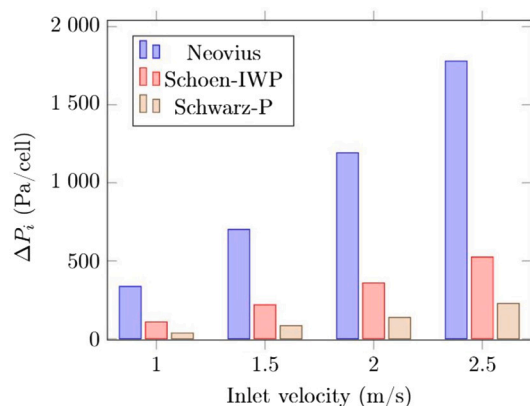


Fig. 12. Pressure drop for different cell geometries with a cell size of 5 mm, $\epsilon = 0.35$.

The study was conducted experimentally, numerically and analytically. The pressure drop and fluid flow details acquired by different methods are compared and discussed in detail.

3.1. Flow dynamics

The numerical modeling of fluid flow inside Schoen's I-WP, Neovius and Schwarz-P TPMS geometries are shown in Fig. 8, Fig. 9 and Fig. 10, respectively. The simulations were performed for the different geometries, cell sizes, inlet velocities and porosity values. Only TPMS geometry had a significant effect on the general flow distribution. Therefore, the results are shown for each TPMS structure with the same set of geometric and boundary conditions.

The flow distribution inside Schoen's I-WP is distinguished by the zones without fluid motion. These zones can be seen in Fig. 8 (b, c, d) to the left and right of the cell center. That flow behavior is the reason for considerable backflow and, therefore, even higher tortuosity of the flow. Authors Attarzadeh et al. (2021) observed a similar effect on other TPMS geometries. To measure the flow tortuosity with the numerical calculations, the velocity field averaging approach was chosen (Graczyk and Matyka, 2020). The tortuosity was calculated as follows:

$$\tau = \frac{\bar{v}}{\bar{v}_x} \quad (14)$$

For the current geometry flow tortuosity was about 5%. In the Schoen's I-WP TPMS packed bed higher fluid motion is observed at the center of each cell. The reason is the fluid collision at the cell center from four directions due to the cell geometry (Fig. 8c). After the collision, the flow splits again and exits the cell. This flow pattern leads to a boundary layer destruction, which enhances the flow turbulent motions. The higher flow turbulence positively affects the heat transfer characteristics, as was noted by several authors (Yan et al., 2023). Also in Fig. 8(d) the vertical motion occurs after the airflow is discharged from the packed bed. This fluid motion causes unfavorable circulations, which affect the flow measurements (Torregrosa et al., 2017).

The flow pattern inside the Neovius TPMS is presented in Fig. 9. The fluid enters the cell through four different openings at the center of each cell side. At the cell outlet the flow collides, providing better fluid mixing, which is favorable in practical applications (Wang et al., 2023; Yeranee and Rao, 2022; Fan et al., 2022; Guo et al., 2023). In the central part of each cell regions backflow and swirled flow are observed. This flow behavior is the reason for the higher pressure drop. The helical motions increase as the ratio of the cross-sectional area of the central part to the inflow part increases. That ratio increases due to the structural porosity change. Due to the backflow and helical motions average tortuosity of 1.1 on Neovius TPMS medium was observed. The fluid mixing in the radial direction between neighboring cells is very low. However, intercellular mixing occurred in the axial direction.

The velocity streamlines inside Schwarz-P TPMS (Fig. 10) differ from Neovius and Schoen's I-WP. The flow enters the Schwarz-P cell through only one face instead of four, as is the case for Neovius and Schoen's I-WP. As a result, that topology prevents intercellular fluid mixing. The one possible channel for the mixing is located perpendicularly to the main flow and imposes high local resistance. Therefore, fluid velocity inside these channels is practically zero. The flow inside the Schwarz-P cell is very similar to a flow in the tube. The flow tortuosity inside the tube is 1, but in the Schwarz-P cell tortuosity is higher by about 3%. The effect is attributed to the flow expansion in the cell center, which increases the fluid travel distance. One should include the tortuosity term to estimate the pressure drop inside the Schwarz-P structure correctly. The helical motions inside the cell and at the outflow from the packed bed are not observed. At the cell outflow air gradually expands until it mixes with air from the other cells.

3.2. Pressure drop

3.2.1. Numerical simulation

Pressure drop along the axial direction for three TPMS packed bed geometries, namely: a) Schwarz-P, b) Neovius and c) Schoen's I-WP is presented in Fig. 11. The pressure drop inside the packed bed occurs due to the constant flow direction changes, fluid acceleration in narrow channels, mixing and frictional forces. In packed beds, the main pressure losses are determined by local resistances (Noël et al., 2023). The local losses have a non-linear behavior as the entrance pressure drop is different for each geometry due to the abrupt change of flow cross-sectional area at the entrance of each cell (Duan and Ma, 2020). For the Schwarz-P geometry the entrance pressure gradient was the highest, while for the Schoen I-WP cell it had the lowest value (Fig. 11).

In the present estimation, pressure drop was calculated by averaging the total pressure drop by the number of cells in the axial direction:

$$\Delta P = \Delta P_i \cdot n \quad (15)$$

In the equation (15) the influence of entrance losses was significantly decreased as it was averaged for the total number of cells. The entrance pressure drop is an essential characteristic of flow inside TPMS packed bed structures, and it should be studied separately in future studies.

The comparison of pressure drop for three geometries of TPMS-based packed beds is presented in Fig. 12. The cell size, porosity and inlet velocity were 5 mm, 0.351 and 1 to 2.5 m/s, respectively. The lowest pressure drop per unit cell was observed on Schwarz-P TPMS, while for Schoen's I-WP it was by 3.3-3.5 times higher and for Neovius by 8.5-10 times. Schwarz-P cell was characterized by the almost straight flow that resembled the flow inside a tube. Schoen's I-WP cell had a channel cross-section higher by two times than that of Schwarz-P, but the flow constantly twisted and collided inside every cell, leading to a higher pressure drop. Neovius TPMS cell had a similar flow pattern to Schoen's I-WP, but the cross-sectional area of the flow channel was 20% lower than that of Schwarz-P.

The implementation of the discussed structures in practical applications depends on the pressure drop profiles. Schwarz-P TPMS structure could be implemented in heat exchangers and filtration systems where minimizing the pressure drop is crucial. Schoen's I-WP TPMS structure could be used in fluidized bed reactors where medium pressure drop is tolerated and efficient fluid-solid contact is important. Neovius structure has the highest pressure drop and may be used in applications where the high pressure drop is acceptable. Due to the high velocity values inside the structure and constant intersections of velocity streamlines, it provides better fluid mixing. In catalytic chemical reactors pressure drop could be negligible compared to the catalyst cost, for example, in steam methane reforming (Pashchenko et al., 2024). Thus, the Neovius structure could be beneficial in catalytic chemical reactions as it will enhance reactants mixing and increase reaction rates.

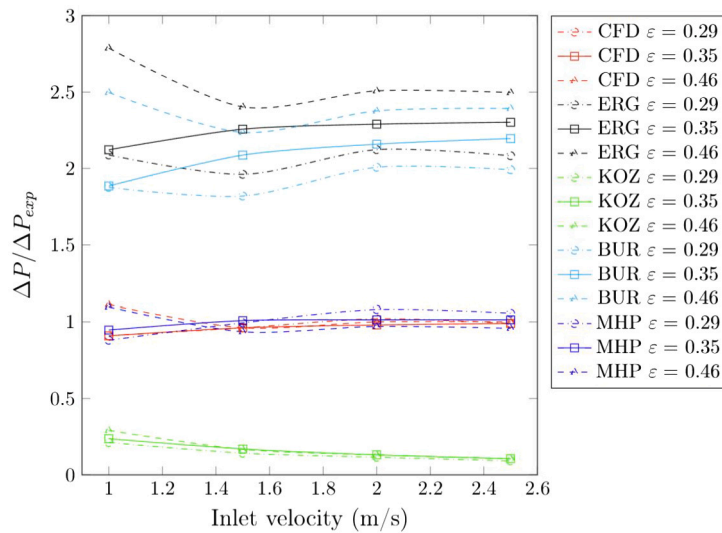


Fig. 13. Analytical solutions for pressure drop per one cell row of Schwarz-P TPMS packed bed from equations: Ergun (eq. (4)), Kozeny-Carman (eq. (6)), Burke-Plummer (eq. (7)), Modified Hagen-Poiseuille (eq. (8)).

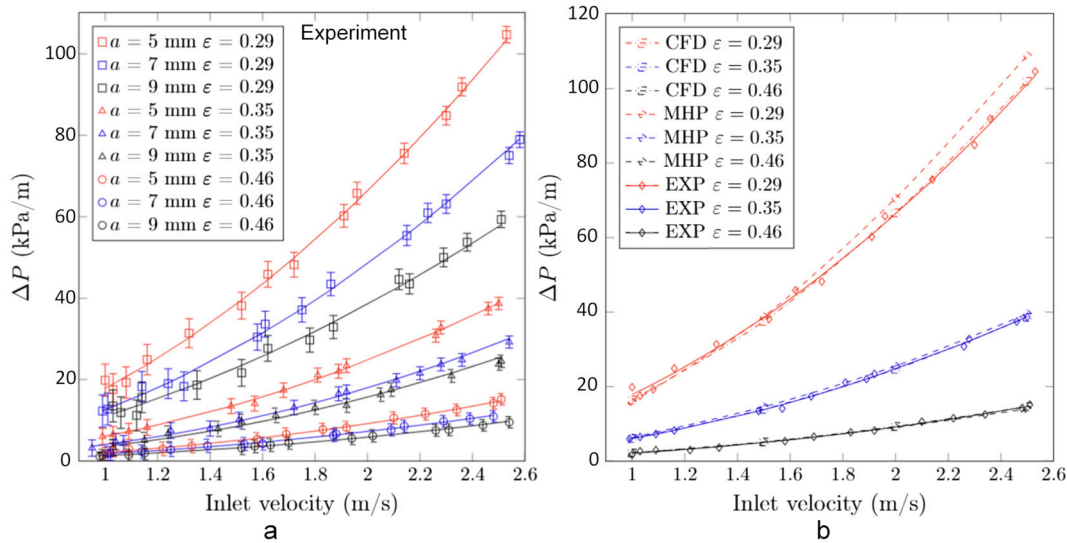


Fig. 14. Pressure drop per unit cell for Schwarz-P TPMS with various porosity and cell size for: a) Experiment b) Comparison between Modified Hagen-Poiseuille Equation (MHP), CFD results (CFD) and Experimental results (EXP) for $a = 7$ mm.

3.2.2. Analytical solution

Several traditional equations describe the pressure drop in a packed bed. The most widely used equations are: Ergun, Burke-Plummer and Kozeny-Carman. The pressure drop predictions of these three equations and a modified Hagen-Poiseuille equation (eq. (14)) for Schwarz-P cell geometry were compared in Fig. 13. The similarity of the Ergun, Burke-Plummer and a modified Hagen-Poiseuille equation is obvious (Fig. 13a, c, d). The similarity shows that turbulence behavior is captured correctly, but the values differ significantly. The modified Hagen-Poiseuille equation includes the effect of the specific tortuosity of the packed bed and pressure loss due to the expansion and contraction of the flow. Therefore, its pressure drop values (Fig. 13d) are lower by 55% than that for the Ergun equation (Fig. 13c) and are closer to the experimental results as is shown in section 3.3. The Kozeny-Carman equation failed to predict pressure loss in the turbulent flow as it was developed for laminar flow.

3.3. Verification

The experimental study was carried out to validate the results of the CFD modeling and analytical solution. The experimental setup is shown

in section 2.2 and described in detail. The pressure drop values were obtained for the packed bed with Schwarz-P cell geometry for nine configurations with various cell sizes (5 mm, 7 mm, 9 mm) and cell porosity (0.29, 0.35, 0.46). The measured pressure drop for all cases is shown in Fig. 14a. An additional line was drawn for each case representing the average values.

The pressure drop values obtained by experiment, CFD and analytical solution are compared in Fig. 14b. Both experimental and CFD pressure drop predictions are in good agreement with a maximum deviation of less than 7.2%. That fact confirms the reliability of the data obtained by CFD modeling. The average deviation between the modified Hagen-Poiseuille equation and experimental results was 3.9%. This deviation is caused by the nature of the equation itself. A modified Hagen-Poiseuille equation was proposed to calculate pressure drop on spherical particles. Schwarz-P particle geometry is different from spherical even though the flow inside such a packed bed strongly resembles the flow inside a packed bed with spherical particles. Schwarz-P particle non-spherical form influence on pressure drop was reduced by introducing the sphericity factor ϕ . However, in order to fully take into account the non-spherical shape of the particle, the equation must be rewritten

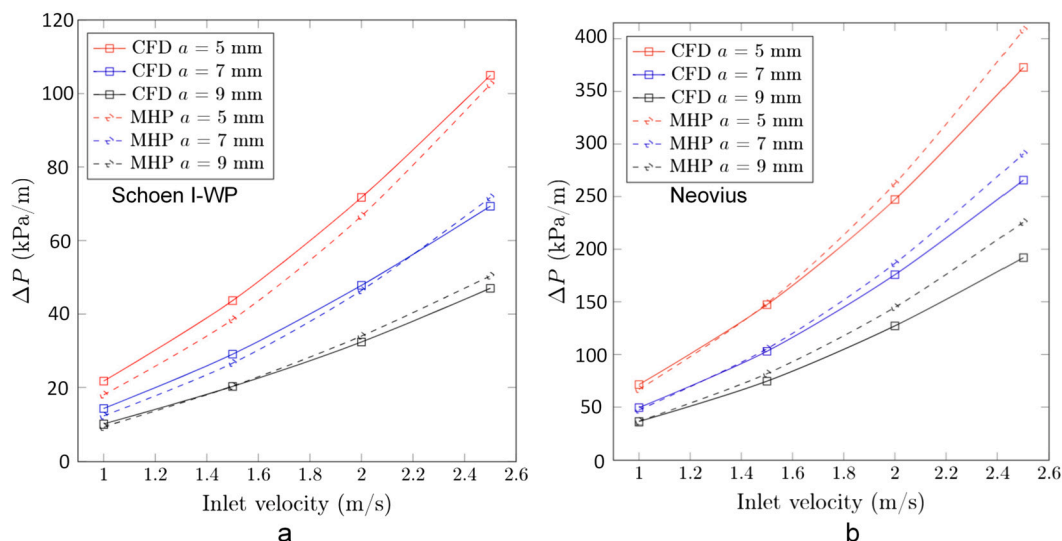


Fig. 15. Comparison of pressure drop calculated by CFD and Modernised Hagen Poiseuille equation (eq. (13)) for: a) Schoen I-WP b) Neovius TPMS.

for the volume of the Schwarz-P TPMS particle. Overall, the modified Hagen-Poiseuille equation can predict pressure drop inside a packed bed with Schwarz-P particle geometry for different cell sizes, structure porosity and flow velocity with high accuracy.

The pressure drop prediction of the modified Hagen-Poiseuille equation (eq. (8)) for Schoen I-WP and Neovius TPMS is presented in Fig. 15. The modified Hagen-Poiseuille equation captures well both the magnitude and behaviors of the pressure drop. However, the deviation between analytical and numerical results is present. For Schoen I-WP TPMS (Fig. 15a) equation overestimates pressure drop for large cells of 9 mm and underestimates for smaller cells of 5 mm. For Neovius TPMS (Fig. 15b) the equation predicts higher pressure drop for all cases. This deviation is attributed to the form of the flow channel. The expansion and contraction is non-linear for Schoen I-WP and Neovius TPMS and therefore influences the local flow resistance. Also a different number of flow channels also greatly affected the pressure drop as it was 4 times higher for Neovius than for Schoen's I-WP TPMS cell.

4. Conclusion

In this study the fluid flow characteristics were studied on large-scale TPMS-based cellular structures on three different cell geometries: Schwarz-P, Neovius, Schoen I-WP. The numerical, experimental and analytical approaches were used to find the equation describing pressure drop on TPMS packed bed under various boundary and geometrical parameters. The main results are the following:

- The classical equations for pressure drop calculations: Ergun and Burke-Plummer overpredicted the pressure drop by about two times and Kozeny-Carman equation underpredicted it by more than ten times compared with experimental data. These equations didn't consider variable tortuosity of the flow and local losses of sudden expansion and contraction of the flow.
- A modified Hagen-Poiseuille equation (eq. (13)) was proposed for pressure drop prediction on Schwarz-P, Neovius and I-WP TPMS structures. The pressure drop predicted by this equation has low discrepancies with experimental and numerical data for $Re_{in} = 1850 - 9250$; cell size $a = 5 - 9$ mm and cell porosity $\varepsilon = 0.29 - 0.46$.
- Pressure drop contours showed that significant losses are at the inlet and outlet of the cell, and linear losses inside the cell are relatively small.
- Pressure drop on Neovius, Schoen I-WP and Schwarz-P TPMS cells was compared. The lowest pressure drop was observed on Schwarz-

P TPMS, while for Schoen's I-WP it was by 3.3-3.5 times higher and for Neovius by 8.5-10 times.

The developed Modified Hagen-Poiseuille equation predicts well the pressure drop on a Schwarz-P, Schoen's I-WP and Neovius TPMS packed bed of various cell configurations. Based on pressure drop performance Schwarz-P TPMS structure could be implemented in heat exchangers and filtration systems; Schoen's I-WP TPMS structure could be used in fluidized bed reactors and the Neovius TPMS structure could be beneficial in catalytic chemical reactions. In future works, equations may also be developed for pressure drop calculation on the other TPMS geometries.

CRedit authorship contribution statement

D. Bragin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **I. Karpilov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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